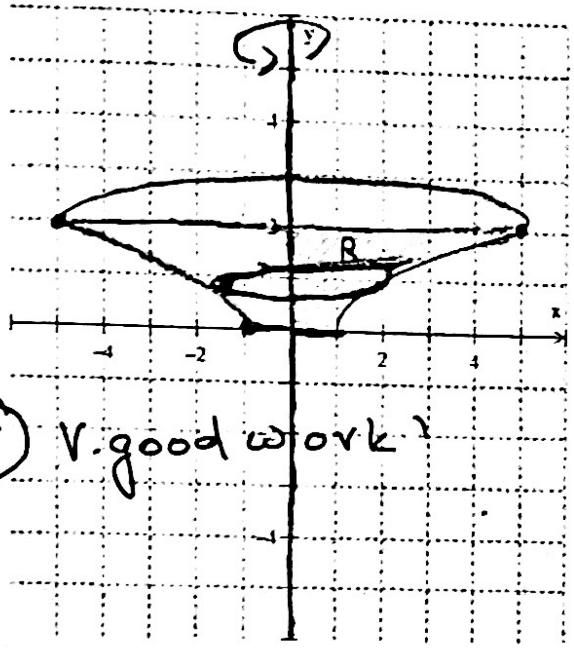
1. [5 pts] Set up the integral to find the volume of the solid obtained by rotating the region bounded by curves $y = \sqrt{x-1},\ y = 2, x = 0.\ and\ y = 0$ about the y-axis. Sketch the region, solid and typical disk/washer. Do not evaluate the integral.



+5 V. good work!

R = right − left

$R = \sqrt{x-1} - 0$

$R = y^2 + 1$

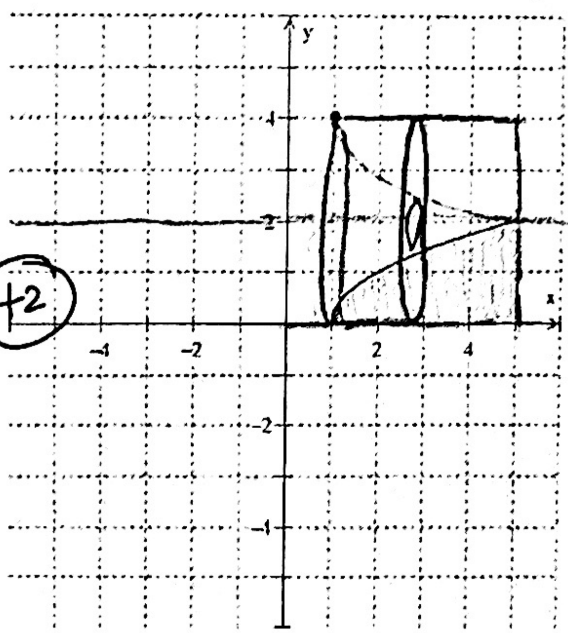
$(y)^2 = (\sqrt{x-1})^2$

$y^2 = x - 1$

$y^2 + 1 = x$

$$\int_0^2 \pi r^2\, dy \Rightarrow \int_0^2 (y^2+1)^2 \pi\, dy$$

2. [5 pts] Set up the integral to find the volume of the solid obtained by rotating the region bounded by curves $y = \sqrt{x-1}, x = 5, y = 0$, about the line $y = 2$. Sketch the region, solid and typical disk/washer. Do not evaluate the integral.



+2

R = top − bottom

$R = 2 - \sqrt{x-1}$

$$\int_0^5 \pi R^2 - \int_0^5 \pi r^2\, dx$$

$$\int_0^5 \pi (2 - \sqrt{x-1})^2\, dx$$

$R = 2 \quad r = 2 - \sqrt{x-1}$

3. [4 pts] If the work required to stretch a spring 0.5 ft beyond its natural length is 6 $ft - lb$, how much work is needed to stretch it 3 in beyond its natural length. Evaluate the Integral and write final answer with the correct unit of measurements.

$w = f \cdot d$

1 ft = 12 inches

$0.5 \text{ ft} \cdot \frac{12 \text{ inches}}{1 \text{ ft}} =$

$\int_0^6 kx \Rightarrow k\left[\frac{x^2}{2}\right]_0^6 = k(18)$

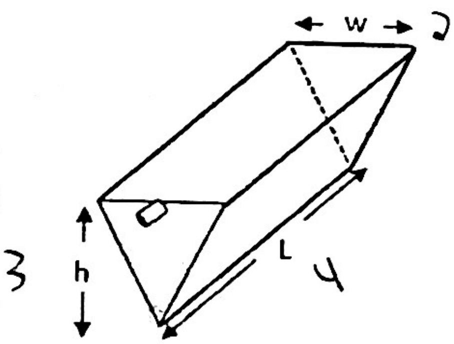
+4

$6 = k(18) = \frac{1}{3}$

V. good work!

$\int_0^3 \frac{1}{3}x \Rightarrow \frac{1}{3}\left[\frac{x^2}{2}\right]_0^3 = \boxed{1.5 \text{ ft-lb}}$

4. [6 points] A tank is full of kerosine. Set up the integral that represent the work required to pump the kerosine out of the spout. Use the fact that density of kerosine is 820 kg/m³. Assume w= 2 m, L = 4 m, and h = 3 m. Clearly define the variable and do not evaluate the integral. Write the correct unit of measurement.



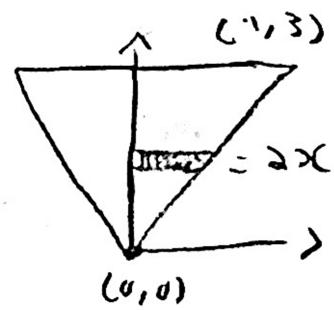
$F = m \cdot g \qquad m = d \cdot v$

$F = (3280)(w\,dy) \cdot 9.8 \quad (820)\, 4 \cdot w\, dy$

$d = (3-y)$

+6

$\int_0^3 32144 \cdot w \cdot (3-y)\, dy$



y = consider a layer y from the bottom

(1,3) (0,0)
$x_1 y_1 \quad x_2 y_2$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-3}{0-1} = 3 \quad m = 3$

$\boxed{\int_0^3 32144\left(\frac{2y}{3}\right)(3-y)\, dy \text{ J}}$

$y - y_1 = m(x - x_1) = y - 3 = 3(x-1)$

$y - 3 = 3x - 3$

$y = 3x$

$x = \frac{y}{3} \Rightarrow \frac{2y}{3}$

V. good work!

ow
rite

5. [5 points] A 240 lb cable that is 60 ft long hangs vertically from the top of a tall building. How much work is done is lifting ¼ the rope to the top of the building. Clearly define the variable and evaluate the integral. Write the correct unit of measurement.

$F = \left(\frac{240}{60}\right)$ ✓ $\qquad d = (60-y)$

$$\int_{45}^{60} 4(60-y)\,dy + \int_{0}^{45} 4(15)\,dy$$

+4

Numericalerror!

$$4\left[60y - \frac{y^2}{2}\right]_{45}^{60} + \left[60y\right]_{0}^{40}$$

$$4\left[60(60) - \frac{(60)^2}{2}\right] - 4\left[60(45) - \frac{(45)^2}{2}\right] + \left[60(40)\right]$$

$$4(3600 - 1800) - 4(2700 - 1012.5) + 2400 = \boxed{2850 \text{ ft-lb}}$$

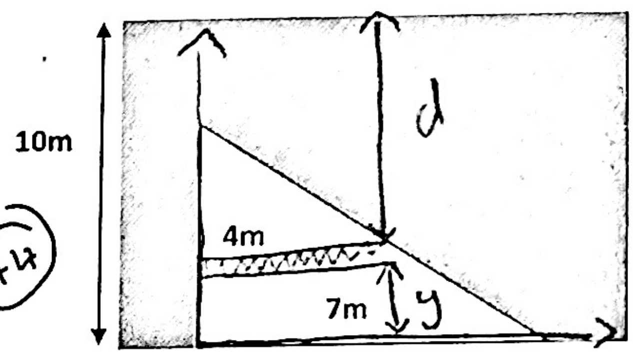
6. [5 points] A triangular vertical plate is submerged in water and as see in the picture below. Write the integral that express the hydrostatic force experienced by the plate. Water density is 1000 kg/m³. Clearly define the variable and do not evaluate the integral. Write the correct unit of measurement.

$F = P \cdot A \qquad P = \rho \cdot g \cdot d$

$= (1000)(9.8)(10-y)$

$A = l \cdot w$

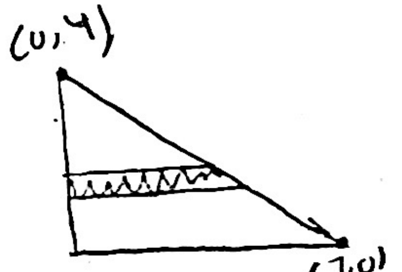
$A = w \cdot \Delta y$



+4

$$\int_0^4 9800\,(10-y)\,w \cdot dy$$

y = consider a layer
at height y from ✓
the bottom



$y - y_1 = m(x - x_1)$

$y - 4 = -\frac{4}{7}(x - 0)$

$y - 4 = -\frac{4}{7}x$

$\frac{-7y + 28}{4}$

$x_1 \; y_1 \qquad x_2 \; y_2$

$(0, 4) \qquad (7, 0)$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-4}{7-0} = -\frac{4}{7}$

$$\boxed{\int_0^4 9800\,(10-y)\left(\frac{-7y+28}{4}\right)dy \quad \text{Newtons}}$$

Pascals ✓

[5 pts] Find the arc length of the curve: $y = \ln(\sin x)$, $\frac{\pi}{4} \le x \le \pi/2$. Set up the integral and evaluate the integral.

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$y = \mathrm{LN}(\sin(x))$$

$$\frac{dy}{dx} = \frac{\cos(x)}{\sin(x)} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{\cos(x)}{\sin(x)}\right)^2 \Rightarrow \frac{\cos^2(x)}{\sin^2(x)} + 1$$

$$\frac{\cos^2(x) + \sin^2(x)}{\sin^2(x)} \Rightarrow \frac{\sin^2(x) + \cos^2(x)}{\sin^2(x)} \Rightarrow \frac{1}{\sin^2(x)}$$

$$\csc^2(x) \Rightarrow \sqrt{\csc^2(x)} \Rightarrow \csc(x)$$

$$-\ln|\csc x + \cot x| + C$$

$$\int_{\pi/4}^{\pi/2} \csc(x) \Rightarrow \left[-\csc(x)\cot(x)\right]_{\pi/4}^{\pi/2}$$

+4

$$\left[-\csc\left(\tfrac{\pi}{2}\right)\cot\left(\tfrac{\pi}{2}\right)\right] - \left[-\csc\left(\tfrac{\pi}{4}\right)\cot\left(\tfrac{\pi}{4}\right)\right]$$

$$0 \qquad -(-\sqrt{2}) = \boxed{\sqrt{2}}$$

[5 pts] Solve the differential equation $3x^2\frac{dy}{dx} - 4 = 5y$ using the variable separable method.

$$3x^2\frac{dy}{dx} - 4 = 5y \Rightarrow \frac{dy}{dx} = \frac{(5y+4)}{3x^2}$$

+4

$$\frac{dy}{dx} = \frac{1}{3x^2}\cdot(5y+4) \Rightarrow \frac{1}{(5y+4)}\cdot dy = \frac{1}{3x^2}\,dx$$

$$\int \frac{1}{(5y+4)}\cdot dy = \int 3x^{-2}\,dx \Rightarrow$$

$$u = 5y + 4$$
$$du = 5\,dy$$
$$\frac{du}{5} = dy$$

$$\frac{1}{5}\int \frac{1}{u}\cdot du = \int \cancel{3}^{\,1}_{\,2} x^{-2}\,dx \Rightarrow \boxed{\frac{1}{5}\mathrm{LN}|5y+4| = -\underline{3}x^{-1} + C}$$

3 must be the denominator!

9. [5 pts] Approximate the Integrate $\int_1^2 \sqrt{x^2+3}\,dx$ using trapezoidal rule for $n = 5$ subintervals. Write your final answer correct to 3 decimal places. Clearly state $\Delta x, x_0, x_2, \dots, x_n, f(x)$ values.

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{5} = \frac{1}{5}$$

$x_0 = 1$

$x_1 = \frac{6}{5}$

$x_2 = \frac{7}{5}$

$x_3 = \frac{8}{5}$

$x_4 = \frac{9}{5}$

$x_5 = 2$

$$\frac{1}{10}\left(\sqrt{4} + 2\sqrt{\frac{111}{25}} + 2\sqrt{\frac{124}{25}} + 2\sqrt{\frac{139}{25}} + 2\sqrt{\frac{156}{25}} + \sqrt{7}\right)$$

$$= \boxed{2.303}$$

+5

V. good work!

10. [5 pts] Evaluate the Integrate: $\int \frac{2x+3}{x(x^2+9)}\,dx$

$$\int \frac{2x+3}{x(x^2+9)} \Rightarrow \int \frac{A}{x} + \frac{Bx+C}{(x^2+9)} \Rightarrow \int \frac{A}{x} + \frac{Bx}{(x^2+9)} + \frac{C}{(x^2+9)}$$

$$A\int \frac{1}{x}\,dx \; ; \int \frac{B\cancel{x}}{u} \cdot \frac{du}{2\cancel{x}} + \int \frac{C}{(x^2+9)} \Rightarrow A\ln|x| + \frac{B}{2}\ln|x^2+9| + \left(\frac{C}{3}\right)\tan^{-1}\left(\frac{x}{3}\right)$$

$$\frac{2x+3}{x(x^2+9)} = \frac{A}{x} + \frac{Bx+C}{(x^2+9)} \Rightarrow 2x+3 = A(x^2+9) + (Bx+C)(x)$$

$$2x+3 = Ax^2 + 9A + Bx^2 + Cx \Rightarrow A + B = 0 \Rightarrow \frac{1}{3} + B = 0 = B = -\frac{1}{3}$$

$C = 2$

$9A = 3 = A = \frac{1}{3}$

$C = 2$

$A = \frac{1}{3}$

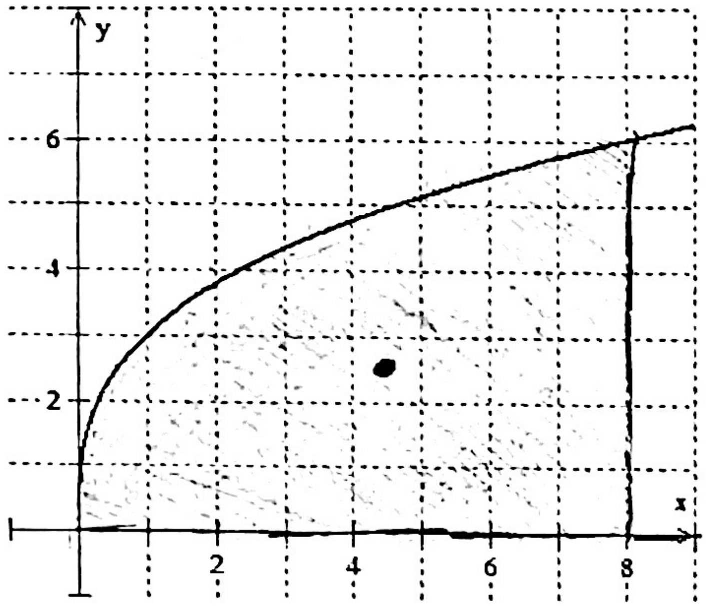
$B = -\frac{1}{3}$

$$\boxed{\frac{1}{3}\ln|x| - \frac{3}{2}\ln|x^2+9| + \left(\frac{2}{3}\right)\tan^{-1}\left(\frac{x}{3}\right) + C}$$

$$-\frac{1}{3} \times \frac{1}{2} = -\frac{1}{6}$$

+4.5

11. [6 points] Sketch the region bounded by the curves $y = 3x^{1/3}, y = 0, and\ x = 8$ and find the exact location of the centroid and plot it.



$A = \int_a^b F(x)\,dx$

$\int_0^8 3x^{1/3} \Rightarrow \left[\frac{9}{4}x^{4/3}\right]$

$= \boxed{36}$

$\bar{x} = \frac{1}{A}\int_a^b x\,[F(x)]\,dx$

$\frac{1}{36}\int_0^8 3x^{4/3}\,dx$

$\frac{1}{36}\left[\frac{9}{7}x^{7/3}\right]_0^8$

$= \boxed{\frac{32}{7}}$

$\bar{y} = \frac{1}{A}\int_a^b \frac{1}{2}[F(x)]^2\,dx$

$\frac{1}{36}\int_0^8 \frac{1}{2}(3x^{1/3})^2\,dx$

$\frac{1}{36}\int_a^8 \frac{1}{2}(9x^{2/3})\,dx$

$\frac{1}{36}\int_0^8 \left(\frac{9}{2}x^{2/3}\right)dx$

$\frac{1}{36}\left[\frac{27}{10}x^{5/3}\right]_0^8$

$= \boxed{\frac{12}{5}}$

$(\bar{x}, \bar{y})$

$\left(\frac{32}{7}, \frac{12}{5}\right)$ ✓

V. good work!

+6